The Journal of Open Source Software

DaedalusData: A Dockerized Tool for Exploration, Knowledge Externalization and Labeling of Image Collections

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DOI: 10.xxxxxx/draft

Software

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Submitted: 20 March 2025
Published: unpublished

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Summary

DaedalusData is an open-source tool for exploring and labeling collections of 2D images. It shows a collection as a two-dimensional projection of image thumbnails, computed from image features, metadata or both, and lets users filter, select and label images in the same web interface. It is meant for the early phase of an analysis, when the categories that matter have not been decided yet and the labels are the result of the work. Users build several sets of labels, called label alphabets, while looking at the data, and feed them back into the projection. DaedalusData runs locally with Docker Compose, so the data stays on the user's machine.

Statement of need

Many scientific image collections come without a taxonomy. Before a classifier can be trained or an annotation campaign planned, someone has to look through thousands of images, find the patterns that recur and give them names. That person is usually a domain expert who can run a Jupyter notebook but should not have to build or host a web application to do this work. Existing tools split this work in two: some project a collection for visual inspection, others apply a label schema that already exists. Moving between them means exporting, scripting and re-importing data.

DaedalusData's workflow puts both steps into one loop. Users explore a projection, select groups of images with a lasso or metadata filters, assign them to labels, and recompute the projection with those labels included. Several label alphabets can describe the same collection from different perspectives. The target users are researchers who have to categorize an image collection before they know which categories it contains.

State of the field

DaedalusData's approach follows the visual-interactive labeling process (Bernard, Zeppelzauer, et al., 2018), in which the user rather than a model chooses what to label next; Bernard, Hutter, et al. (2018) found this competitive with active learning when few labels exist. Several open-source tools cover parts of this workflow, each from a different starting point. Table 1 compares them with DaedalusData.

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Table 1: Open-source tools for exploring and labeling images.

Tool	Works on	Projection	Labeling	Labels change the projection
Fiji with plugins	Stacks, folders, regions	Separate plugin	Separate plugin	No
napari-clusters-plotter	Objects in a segmented image	Yes	Cluster annotation	No
PixPlot	Image collections	Yes	No	No
CVAT, Label Studio	Single images	No	Yes	No
FiftyOne	Image datasets	Yes	Yes	Not built in
DaedalusData	2D image collections	Yes	Yes	Yes

In the ImageJ ecosystem, one plugin projects an image stack, a folder of images or a results table into a scatter plot, and other plugins assign categories to images or regions of interest, but the two do not share state, so a category assigned in one can neither be seen in the other nor used in the projection (Schindelin et al., 2012). napari-clusters-plotter (Zigutyte et al., 2025) offers UMAP and t-SNE with lasso selection, but for measurements of objects segmented in one image rather than for a collection of images. ilastik (Berg et al., 2019) and QuPath (Bankhead et al., 2017) train pixel and object classifiers and assume that the classes are known. PixPlot (Yale Digital Humanities Lab, 2017) lays out image thumbnails with UMAP but has no labeling. CVAT (CVAT.ai Corporation, 2023) and Label Studio (Tkachenko et al., 2020) apply an existing label schema image by image and show no overview of the collection.

FiftyOne (Moore & Corso, 2020) is the closest in scope. It combines a sample grid, an embeddings panel with lasso selection, and tagging and annotation in one application. It is built for curating machine learning datasets: users typically bring a label schema and a model, and load data and compute embeddings in Python. Its built-in UMAP, t-SNE and PCA layouts are computed from the embeddings alone; labels color and filter the plot but do not change the layout. A label-informed layout requires computing modified embeddings or the points themselves in Python and passing them in. DaedalusData addresses the step before, when the label alphabets are still being developed and each revision can reshape the layout.

Software design

DaedalusData runs as one container with two parts: a Nuxt web application for exploration and labeling, and a Jupyter server with three notebooks that compute its inputs. All data is kept as plain files in a mounted `data/` directory: images, a metadata JSON file, features, projections, and label alphabets as JSON. Other tools can read and write these files directly.

The notebooks do the computation. The first loads the demo dataset; for another collection, the images and a metadata JSON file are copied into `data/` instead. The second computes a ResNet50 embedding (He et al., 2016) for each image and encodes the metadata as a table. The third computes UMAP projections (McInnes et al., 2018) of the image features, of image and metadata features combined, and, for each label alphabet, of the image features with the labels appended as one-hot columns. The notebooks are run as provided, so no code has to be written; changing the feature extractor or the projection method means editing Python in a notebook.

The web interface does everything else (Figure 1). It renders a projection as a field of thumbnails from an image atlas, supports zooming, metadata filters and lasso selection, and saves label

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assignments to disk immediately. Bar charts and violin plots summarize the metadata. To see how new labels change the layout, users re-run the projection notebook and reload the analysis view.

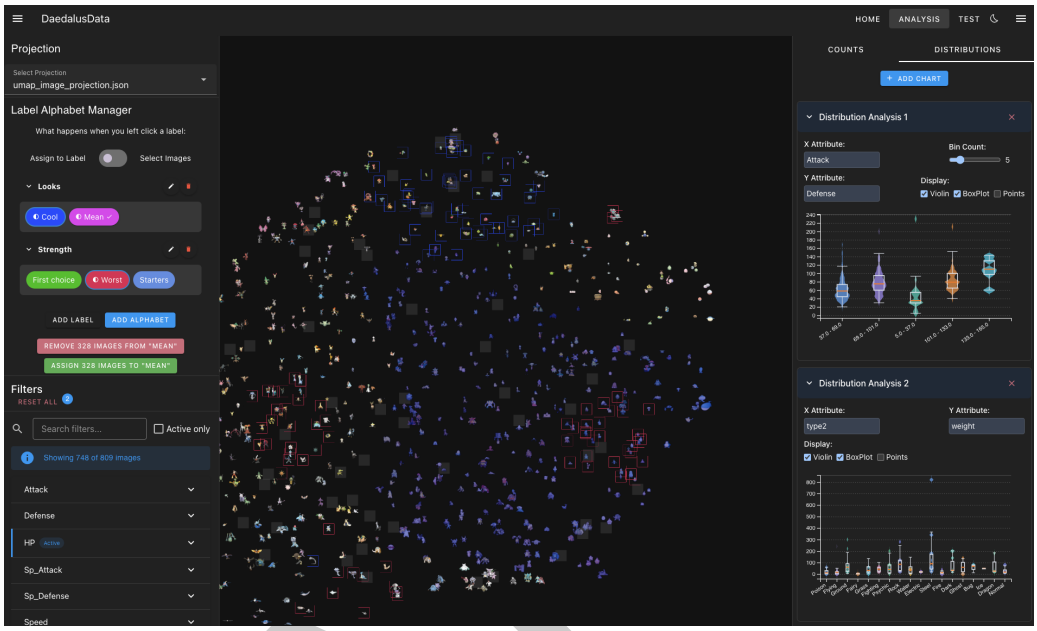


Figure 1: The analysis view on the 809-image demo dataset. Left: projection selector, two label alphabets and metadata filters. Center: the projection, with labeled images framed in the color of their label. Right: violin and box plots of metadata attributes.

Research use

DaedalusData was developed in a design study with quality-control experts who investigate particle contamination in in-vitro diagnostics consumables. They used it to explore and label thousands of 2D particle images and to externalize their knowledge through label-informed projections; the case study and user study are reported in Wyss et al. (2025).

Acknowledgments

This work builds upon research originally published in IEEE Transactions on Visualization and Computer Graphics (Wyss et al., 2025).

References

Bankhead, P., Loughrey, M. B., Fernández, J. A., Dombrowski, Y., McArt, D. G., Dunne, P. D., McQuaid, S., Gray, R. T., Murray, L. J., Coleman, H. G., James, J. A., Salto-Tellez, M., & Hamilton, P. W. (2017). QuPath: Open source software for digital pathology image analysis. *Scientific Reports*, *7*, 16878. https://doi.org/10.1038/s41598-017-17204-5

Berg, S., Kutra, D., Kroeger, T., Straehle, C. N., Kausler, B. X., Haubold, C., Schiegg, M., Ales, J., Beier, T., Rudy, M., Eren, K., Cervantes, J. I., Xu, B., Beuttenmueller, F., Wolny, A., Zhang, C., Koethe, U., Hamprecht, F. A., & Kreshuk, A. (2019). ilastik: Interactive machine learning for (bio)image analysis. *Nature Methods*, *16*, 1226–1232. https://doi.org/10.1038/s41592-019-0582-9

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Bernard, J., Hutter, M., Zeppelzauer, M., Fellner, D., & Sedlmair, M. (2018). Comparing visual-interactive labeling with active learning: An experimental study. *IEEE Transactions on Visualization and Computer Graphics*, *24*(1), 298–308. https://doi.org/10.1109/TVCG.2017.2744818

Bernard, J., Zeppelzauer, M., Sedlmair, M., & Aigner, W. (2018). VIAL: A unified process for visual interactive labeling. *The Visual Computer*, *34*(9), 1189–1207. https://doi.org/10.1007/s00371-018-1500-3

CVAT.ai Corporation. (2023). *Computer vision annotation tool (CVAT)*. Zenodo. https://doi.org/10.5281/zenodo.4009388

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–778. https://doi.org/10.1109/CVPR.2016.90

McInnes, L., Healy, J., Saul, N., & Großberger, L. (2018). UMAP: Uniform manifold approximation and projection. *Journal of Open Source Software*, *3*(29), 861. https://doi.org/10.21105/joss.00861

Moore, B. E., & Corso, J. J. (2020). *FiftyOne*. https://github.com/voxel51/fiftyone

Schindelin, J., Arganda-Carreras, I., Frise, E., Kaynig, V., Longair, M., Pietzsch, T., Preibisch, S., Rueden, C., Saalfeld, S., Schmid, B., Tinevez, J.-Y., White, D. J., Hartenstein, V., Eliceiri, K., Tomancak, P., & Cardona, A. (2012). Fiji: An open-source platform for biological-image analysis. *Nature Methods*, *9*(7), 676–682. https://doi.org/10.1038/nmeth.2019

Tkachenko, M., Malyuk, M., Holmanyuk, A., & Liubimov, N. (2020). *Label Studio: Data labeling software*. https://github.com/HumanSignal/label-studio

Wyss, A., Morgenshtern, G., Hirsch-Hüsler, A., & Bernard, J. (2025). DaedalusData: Exploration, knowledge externalization and labeling of particles in medical manufacturing — a design study. *IEEE Transactions on Visualization and Computer Graphics*, *31*(1), 54–64. https://doi.org/10.1109/TVCG.2024.3456329

Yale Digital Humanities Lab. (2017). *PixPlot: A WebGL viewer for UMAP or t-SNE-clustered images*. https://github.com/YaleDHLab/pix-plot

Zigutyte, L., Savill, R., Müller, J., Zoccoler, M., Wagner, T., & Haase, R. (2025). *napari-clusters-plotter*. Zenodo. https://doi.org/10.5281/zenodo.5884657

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